

## Project Planning Phase

### Project Planning Template (Product Backlog, Sprint Planning, Stories, Story points)

|               |                                                                                     |
|---------------|-------------------------------------------------------------------------------------|
| Date          | 29June 2026                                                                         |
| Team ID       |                                                                                     |
| Project Name  | <b>Plugging into the Future: An Exploration of Electricity Consumption Patterns</b> |
| Maximum Marks | 5 Marks                                                                             |

#### Product Backlog, Sprint Schedule, and Estimation (4 Marks)

| Sprint   | Functional Requirement (Epic) | User Story Number | User Story / Task                                   | Story Points | Priority | Team Members |
|----------|-------------------------------|-------------------|-----------------------------------------------------|--------------|----------|--------------|
| Sprint 1 | Data Collection               | USN1              | Gather electricity consumption dataset (2019–2020)  | 2            | Medium   | Member 1     |
| Sprint 1 | Data Collection               | USN2              | Load dataset into Tableau/Excel                     | 1            | Low      | Member 2     |
| Sprint 1 | Data Preparation              | USN3              | Handle missing values                               | 3            | High     | Member 1     |
| Sprint 1 | Data Preparation              | USN4              | Create calculated fields and date fields            | 3            | High     | Member 2     |
| Sprint 1 | Data Preparation              | USN5              | Handle inconsistent and duplicate data              | 3            | High     | Member 1     |
| Sprint 2 | Data Visualization            | USN6              | Create line chart for consumption trends            | 2            | Medium   | Member 2     |
| Sprint 2 | Data Visualization            | USN7              | Create bar charts for state and regional comparison | 2            | Medium   | Member 1     |
| Sprint 2 | Data Visualization            | USN8              | Create map visualization for state-wise consumption | 5            | High     | Member 2     |
| Sprint 2 | Dashboard Development         | USN9              | Develop interactive dashboard                       | 5            | High     | Member 1     |
| Sprint 2 | Reporting                     | USN10             | Generate reports and project story                  | 5            | High     | Member 2     |

**Project Tracker, Velocity & Burndown Chart: (4 Marks)**

| Sprint   | Total Story Points | Duration | Sprint Start Date | Sprint End Date (Planned) | Story Points Completed (as on Planned End Date) | Sprint Release Date (Actual) |
|----------|--------------------|----------|-------------------|---------------------------|-------------------------------------------------|------------------------------|
| Sprint 1 | 12                 | 4 Days   | 29-06-2026        | 02-07-2026                | 12                                              | 02-07-2026                   |
| Sprint 2 | 19                 | 5 Days   | 03-07-2026        | 07-07-2026                | 19                                              | 07-07-2026                   |

**Velocity:**

Imagine we have a 10-day sprint duration, and the velocity of the team is 20 (points per sprint). Let's calculate the team's average velocity (AV) per iteration unit (story points per day)

$$AV = \frac{\text{sprint duration}}{\text{velocity}} = \frac{20}{10} = 2$$

**Burndown Chart:**

A burn down chart is a graphical representation of work left to do versus time. It is often used in agile software development methodologies such as Scrum. However, burn down charts can be applied to any project containing measurable progress over time.

<https://www.visual-paradigm.com/scrum/scrum-burndown-chart/>

<https://www.atlassian.com/agile/tutorials/burndown-charts>

**Reference:**

<https://www.atlassian.com/agile/project-management>

<https://www.atlassian.com/agile/tutorials/how-to-do-scrum-with-jira-software>

<https://www.atlassian.com/agile/tutorials/epics>

<https://www.atlassian.com/agile/tutorials/sprints>

<https://www.atlassian.com/agile/project-management/estimation>

<https://www.atlassian.com/agile/tutorials/burndown-charts>